Why changing the load resistance did not change the damping: an open-circuit null result for electromagnetic braking of a magnet–spring oscillator, measured with a low-cost data-acquisition system

Hassan Tariq Malik*[1], Hardyesh Kumar[1], Muhammad Umer Farooq[1], M. Sanan Qureshi[1,†], Sadia Manzoor[2,†], and Pervez Hoodbhoy[1,†]

[1]The Black Hole Science Lab, Islamabad, Pakistan
[2]Department of Physics, COMSATS University Islamabad, Pakistan
[†]Supervisor

Abstract

A permanent magnet oscillating on a spring near a coil is the textbook demonstration of electromagnetic (eddy-current) damping. Theory predicts that the electromagnetic part of the damping coefficient scales as $b_{\mathrm{em}} \propto (N\,\mathrm{d}\Phi/\mathrm{d}z)^2/R_{\mathrm{loop}}$, so it should fall as the resistance in the coil circuit rises. We measured the decay of a 105 g NdFeB magnet on two springs (2.63 Hz and 2.33 Hz) above three hand-wound coils, varying an external series resistance from 0 Ω to 2.2 MΩ. The coil EMF was recorded with a purpose-built data-acquisition system (DAQ) costing about US$43 (RP2040 microcontroller, OPA2134 amplifier, gain 35, noise floor ±34 µV). The decay rate did not depend on resistance: for spring 1, $\gamma = (17.8 \pm 0.6) \times 10^{-3}\,\mathrm{s}^{-1}$ (mean ± s.d., $n = 8$), $Q = 465 \pm 15$, across 15 Ω–9.6 kΩ, and a run at 2.2 MΩ gave the same value. A fit of $\gamma = \gamma_0 + c/(R_c + R_{\mathrm{ext}})$ gives $c = -0.009 \pm 0.008\,\Omega\,\mathrm{s}^{-1}$, which excludes the value $c = +0.019\,\Omega\,\mathrm{s}^{-1}$ predicted from the measured EMF for a loop closed through the coil and external resistor at the 3.7σ level. We trace this to the measurement circuit itself: the coil was closed only through the DAQ's 1 MΩ input bias resistor, which limited the induced current to tens of nanoamperes and made b_{em} about 10^{-6} of the measured damping. Mechanical losses therefore dominated. For spring 2 these losses were strongly amplitude-dependent ($r = 0.98$ between decay rate and initial EMF amplitude). We report the coil-geometry dependence of the induced EMF, give a corrected circuit topology, and estimate that even an ideally closed loop with this coil would contribute only ≈ 7% of the total damping. Design files, firmware, raw data and analysis scripts are open.

1 Introduction

When a magnet moves relative to a conducting loop, the changing flux induces an EMF; the resulting current dissipates energy as Joule heat and, by Lenz's law, exerts a force that opposes the motion. For a magnet oscillating on a spring near a coil, this adds a velocity-proportional term to the equation of motion and appears experimentally as an increase in the exponential decay rate of the oscillation. Quantitative versions of this experiment, with the coil EMF logged by a computer interface, are well established in the teaching literature [1, 2], and related geometries—a magnet falling through a conducting pipe—have been analysed in detail [3–5].

A standard way to show that the extra damping is electromagnetic is to change the resistance of the coil circuit: the dissipated power, and hence b_{em}, scales as $1/R_{\mathrm{loop}}$. In this work, carried out during the 2026 Black Hole Science Internship Programme, we attempted that measurement using

*Corresponding author: clasherpro943@gmail.com

a data-acquisition system built from commodity parts because no digital oscilloscope was available. The DAQ worked well, and it resolves the decay rate of a single run to about 1%. However, the damping did not change with resistance.

This paper reports that null result and explains it quantitatively. Our contributions are: (i) an open, low-cost DAQ for millivolt-level coil EMF with a measured noise floor of $\pm 34\,\mu\mathrm{V}$; (ii) a reanalysis of every recorded run with a single, reproducible decay-fitting procedure; (iii) a quantitative prediction, built from the measured EMF alone, of how large b_{em} *should* have been, and a statistical test showing it was absent; (iv) identification of the cause—a high-impedance input that left the coil effectively open—and a corrected design; (v) evidence that the remaining (mechanical) damping is amplitude-dependent for one of the springs. We think the experiment is useful to others for exactly this reason: measuring the voltage on a coil is not the same as letting current flow in it, and the difference is easy to miss.

2 Theory

2.1 Damped oscillator

With a linear damping force $-b\dot{z}$, the displacement of a mass m on a spring of constant k obeys $m\ddot{z} + b\dot{z} + kz = 0$. In the underdamped case this gives

$$z(t) = A_0\, e^{-\gamma t}\cos(\omega_1 t - \phi), \qquad \gamma = \frac{b}{2m}, \qquad \omega_1 = \sqrt{\omega_0^2 - \gamma^2}, \qquad \omega_0 = \sqrt{k/m}. \tag{1}$$

The stored energy decays as $E(t) = E_0 e^{-2\gamma t}$. The quality factor, defined as 2π times the stored energy divided by the energy lost per cycle, is $Q = \omega_0/(2\gamma) \approx \pi f/\gamma$ for $\gamma \ll \omega_0$. In our data $\gamma/\omega_0 \sim 10^{-3}$, so ω_1 and ω_0 are indistinguishable.

The total decay rate is the sum of independent loss channels,

$$\gamma = \gamma_{\mathrm{mech}} + \gamma_{\mathrm{em}}, \tag{2}$$

where γ_{mech} collects air drag, internal friction in the spring and losses at the suspension, and γ_{em} is the electromagnetic contribution.

2.2 Induced EMF and electromagnetic damping

Let $\Phi(z)$ be the flux of the magnet through one turn of an N-turn coil when the magnet is at axial position z. By Faraday's law and the chain rule,

$$\varepsilon(t) = -N\frac{\mathrm{d}\Phi}{\mathrm{d}t} = -N\frac{\mathrm{d}\Phi}{\mathrm{d}z}\,\dot{z} \equiv -\kappa(z)\,\dot{z}, \tag{3}$$

where $\kappa(z) = N\,\mathrm{d}\Phi/\mathrm{d}z$ is the coupling (units $\mathrm{V\,s\,m^{-1}}$). For a point dipole m_d on the axis of a loop of radius a, $\Phi(z) = \mu_0 m_d a^2/[2(z^2+a^2)^{3/2}]$ [6], so

$$\frac{\mathrm{d}\Phi}{\mathrm{d}z} = -\frac{3\mu_0 m_d a^2 z}{2(z^2+a^2)^{5/2}}, \tag{4}$$

which is zero at the loop's centre and largest in magnitude at $z = \pm a/2$. If the coil circuit is closed through a total resistance R_{loop}, and the coil inductance is negligible ($\omega L \ll R_{\mathrm{loop}}$, well satisfied at 2.6 Hz), the current is $I = \varepsilon/R_{\mathrm{loop}}$. Equating the mechanical work against the braking force to the Joule power gives the position-dependent coefficient

$$b_{\mathrm{em}}(z) = \frac{\kappa(z)^2}{R_{\mathrm{loop}}}. \tag{5}$$

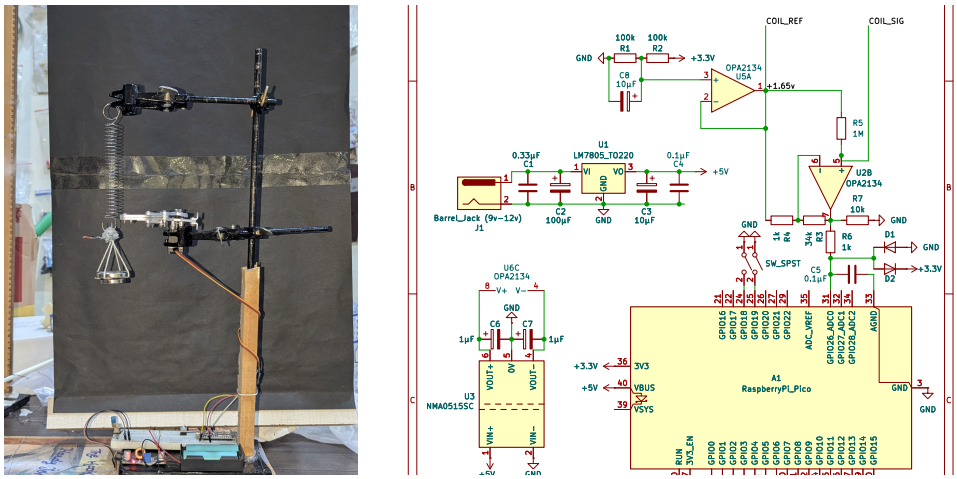


Figure 1: Left: oscillator (two stacked NdFeB magnets in a string cradle) and servo release mechanism; the coil sits below the cradle and is not shown. Right: DAQ schematic. The coil connects between `COIL_SIG` and `COIL_REF`. Note R5 (1 MΩ): it is the only path that closes the coil circuit, which is the origin of the null result discussed in Sec. 5.

Equation (5) needs $\kappa(z)$, which in turn needs the magnet's dipole moment and the coil geometry. We avoid both by using the measured EMF directly. Averaged over a cycle, the Joule power is $\langle P\rangle = \langle\varepsilon^2\rangle/R_{\text{loop}}$. Since $\mathrm{d}E/\mathrm{d}t = -2\gamma E$ and $E = \frac{1}{2}m\omega_0^2 A^2$,

$$\gamma_{\text{em}} = \frac{\langle\varepsilon^2\rangle}{R_{\text{loop}}\, m\, \omega_0^2 A^2} \equiv \frac{c}{R_{\text{loop}}}, \tag{6}$$

where $\langle\varepsilon^2\rangle$ and A are evaluated at the same moment. With a high-impedance voltmeter, ε is the open-circuit EMF, so Eq. (6) predicts γ_{em} for *any* chosen R_{loop} from a single measurement.

3 Methods

3.1 Mechanical apparatus

The oscillator (Fig. 1) consisted of two NdFeB disc magnets stuck together and carried in a string cradle hung from a helical spring. The total oscillating mass was 105 g. Two springs were used, with static spring constants $k_1 = 16.578\,\text{N/m}$ and $k_2 = 19.438\,\text{N/m}$ [method used to measure k, and uncertainty]. A servo-actuated aluminium claw, driven by a separate ESP32 board, released the cradle from rest at a fixed height to standardise the initial displacement. [magnet diameter and thickness; grade; equilibrium height of the magnet relative to the coil plane.]

Three coils were hand-wound with enamelled copper wire on card formers. Coil 1 had a DC resistance $R_c = 15.2\,\Omega$. Coil 2 differed from coil 1 in former diameter, and coil 3 differed from coil 2 in wire gauge [number of turns, mean radius, length, wire gauge and resistance of each coil].

3.2 Data-acquisition system

The coil EMF is tens of millivolts and changes sign. Neither property suits a microcontroller ADC, which reads 0 V to 3.3 V with a step of 0.806 mV (12 bits). The DAQ (Fig. 1, right) addresses both:

- A 100 kΩ/100 kΩ divider, buffered by one half of an OPA2134 (U5A), produces a low-impedance 1.65 V virtual ground (`COIL_REF`). This lets negative EMF be represented within the ADC range.
- The coil signal drives a non-inverting OPA2134 stage (U2B) with gain $G = 1 + 34\,\text{k}\Omega/1\,\text{k}\Omega = 35$. This gives an input range of ±47 mV and an input-referred step of 23 µV. R5 (1 MΩ) sets the DC bias of the input.

- The op-amps run from ±15 V supplied by an isolated NMA0515SC DC–DC converter, so the output never approaches the rails. A 1 kΩ/0.1 µF RC filter and clamp diodes protect the ADC pin.
- The Raspberry Pi Pico (RP2040 [7]) samples at 100 kS/s and averages blocks of samples in RAM (oversampling and decimation). The data reported here were stored at 2 kS/s for up to 60 s and transferred over USB after each run [confirm block size used for these runs: the repository firmware defaults to 100-sample blocks (1 kS/s), but the data files are spaced at 0.5 ms].
- Noise was reduced by battery power, removing the USB 5 V line, short hook-up wiring, local decoupling and a star ground. The measured input-referred noise floor is ±34 µV (signal-to-noise ratio 63 dB at full scale).

The bill of materials totals US$42.57. Schematics, firmware and the bill of materials are in the repository [11].

3.3 Protocol

For spring 1 with coil 1, the external resistance R_{ext} placed in series with the coil was set to 0 Ω, 33.6 Ω, 81.4 Ω, 324 Ω, 1051 Ω, 3180 Ω and 9610 Ω and 2.2 MΩ. The 0 Ω setting was recorded twice. For spring 2 with coil 1 we used 0 Ω, 33.3 Ω, 84.1 Ω, 327 Ω, 997 Ω, 3190 Ω and 9670 Ω, again with 0 Ω repeated. Coils 2 and 3 were recorded once each with $R_{\mathrm{ext}} = 0$ on both springs. Each run lasted about 30 s (about 75 cycles). The cradle's position was also filmed with a phone camera and tracked in software [confirm: Tracker [10], or the custom Python script mentioned in the presentation; frame rate; calibration].

3.4 Analysis

Every run was reanalysed with one script [11]. The raw trace was low-pass filtered with a fourth-order, zero-phase Butterworth filter (cut-off 20 Hz) [8, 9]. The baseline was set to the median EMF in the 0.3 s before release. Peaks of $|\varepsilon|$ were located in each half-cycle, starting at the first peak above half the maximum and keeping peaks above a quarter of the maximum. A straight-line fit to $\ln|\varepsilon_{\mathrm{peak}}|$ versus t gives γ and its standard error. Because $\varepsilon \propto \dot{z}$ for a fixed coupling profile, the EMF envelope decays at the same rate as the displacement. The frequency f was taken from the FFT peak, and $Q = \pi f/\gamma$. Where a damping coefficient is quoted, $b = 2m\gamma$ with $m = 105$ g. Every fit had $R^2 > 0.93$, and 19 of the 21 had $R^2 > 0.97$.

4 Results

4.1 Raw signal

Figure 2 shows a typical run. The EMF is a clean, slightly asymmetric periodic signal at the oscillation frequency, with an exponentially decaying envelope. The asymmetry is expected when the magnet's equilibrium position is offset from the coil plane, since $\kappa(z)$ is then different on the up and down strokes. Unfiltered and filtered traces are almost indistinguishable on this scale: the 34 µV noise floor is about 10^{-3} of the signal.

4.2 Decay rate versus resistance

Table 1 lists every run and Fig. 3 plots γ against $R_c + R_{\mathrm{ext}}$. Neither spring shows any trend. Over the six resistances from 15 Ω to 9.6 kΩ, plus the repeated 0 Ω run, the results are as follows.

- Spring 1: $\gamma = (17.8 \pm 0.6) \times 10^{-3}\,\mathrm{s}^{-1}$, $Q = 465 \pm 15$, $b = (3.74 \pm 0.12)$ g/s.

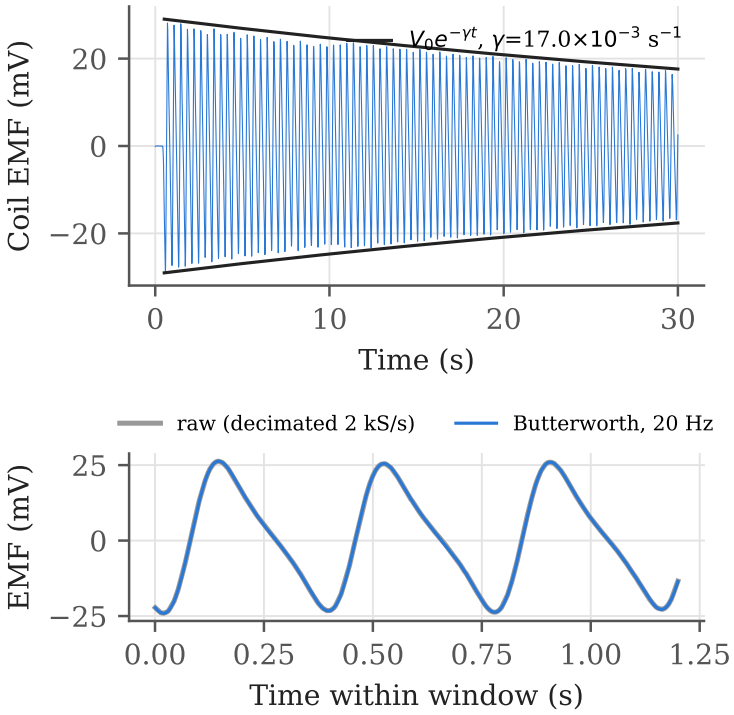


Figure 2: Spring 1, coil 1, $R_{\text{ext}} = 0$. Top: filtered EMF with fitted envelope. Bottom: 1.2-s window, raw (grey, after on-board decimation) and Butterworth-filtered (blue).

- Spring 2: $\gamma = (20.9 \pm 1.9) \times 10^{-3}\,\text{s}^{-1}$, $Q = 353 \pm 33$, $b = (4.40 \pm 0.41)\,\text{g/s}$.

These are means ± standard deviation over $n = 8$ runs. The spread is much larger than the statistical error of each fit ($\approx 1\%$), so it comes from run-to-run irreproducibility, not from the DAQ. The two repeats at $0\,\Omega$ differ by 4% (spring 1) and 21% (spring 2). The $2.2\,\text{M}\Omega$ run on spring 1, where the circuit is effectively open, gives $\gamma = (17.0 \pm 0.2) \times 10^{-3}\,\text{s}^{-1}$. This is indistinguishable from the $15\,\Omega$ runs.

4.3 How large should the electromagnetic damping have been?

For coil 1 on spring 1, the mean rms EMF over the first three cycles was $\varepsilon_{\text{rms}} = 16.2\,\text{mV}$. Taking $A = 2.2\,\text{cm}$, the initial amplitude seen in the video tracks [uncertainty in A], Eq. (6) gives $c = \langle\varepsilon^2\rangle/(m\omega_0^2 A^2) = 0.019\,\Omega\,\text{s}^{-1}$. If the loop had been closed through the coil alone ($R_{\text{loop}} = 15.2\,\Omega$), this would mean $\gamma_{\text{em}} = 1.2 \times 10^{-3}\,\text{s}^{-1}$, about 7% of the measured total and about twice the run-to-run scatter of spring 1. That is small, but it should have been detectable.

To test for it, we fitted $\gamma = \gamma_0 + c/(R_c + R_{\text{ext}})$ to the eight spring-1 runs below $10\,\text{k}\Omega$, weighting each by the run-to-run scatter. The fit gives $c = -0.009 \pm 0.008\,\Omega\,\text{s}^{-1}$. This is consistent with zero and lies 3.7σ below the predicted $+0.019\,\Omega\,\text{s}^{-1}$. For spring 2 the scatter is larger, and the fit ($c = 0.015 \pm 0.026$) cannot distinguish the two cases.

4.4 Coil geometry and the induced EMF

The induced EMF depends clearly on the coil, even though the damping does not (Table 2). With the same magnet and spring, coil 2 (different former diameter) produced about one third of coil 1's EMF, and coil 3 (different gauge) about one ninth. The pattern is the same on both springs. Because $\gamma_{\text{em}} \propto \varepsilon^2$ at fixed R_{loop}, the electromagnetic damping that coils 2 and 3 could have produced is one to two orders of magnitude smaller than coil 1's. Their measured decay rates are, as expected, the same as coil 1's within the run-to-run scatter.

Table 1: Decay rate γ (fit standard error in parentheses), quality factor and initial rms EMF for each run with coil 1. $R_c = 15.2\,\Omega$. The last column gives the damping coefficient from video tracking as reported in the internship report, converted to a rate as $\gamma = b/2m$. Two runs from that report are missing because they were not tracked.

	R_{ext} (Ω)	γ ($10^{-3}\,\text{s}^{-1}$)	Q	ε_{rms} (mV)	γ_{track} ($10^{-3}\,\text{s}^{-1}$)
Spring 1	0 (a)	16.95(22)	488	17.7	11.7
$f = 2.633\,\text{Hz}$	0 (b)	17.71(21)	467	15.3	–
	33.6	17.95(18)	461	15.6	17.3
	81.4	18.22(22)	454	16.6	16.1
	324	18.56(23)	446	17.7	13.3
	1051	17.23(20)	480	15.8	–
	3180	17.40(19)	476	15.9	11.6
	9610	18.44(27)	449	15.3	11.8
	2.2×10^6	16.98(23)	487	4.0	–
Spring 2	0 (a)	24.03(28)	305	18.2	11.6
$f = 2.333\,\text{Hz}$	0 (b)	19.43(18)	377	13.7	–
	33.3	21.64(15)	339	15.0	15.7
	84.1	17.85(20)	411	11.4	13.6
	327	20.74(35)	353	14.1	14.7
	997	22.88(29)	320	17.6	–
	3190	20.01(22)	366	14.1	10.5
	9670	21.00(21)	349	14.9	12.1

Table 2: Coil comparison at $R_{\text{ext}} = 0$. The coil-1 row is the mean over that spring's runs below $10\,\text{k}\Omega$. The predicted γ_{em} assumes a loop closed through $15.2\,\Omega$ and $A = 2.2\,\text{cm}$; the coil-2 and coil-3 resistances are not yet known [measure them].

Spring	Coil	ε_{rms} (mV)	γ ($10^{-3}\,\text{s}^{-1}$)	predicted γ_{em} ($10^{-3}\,\text{s}^{-1}$)
1	1	16.2	17.8 ± 0.6	1.2
1	2	5.9	16.70(42)	0.17
1	3	1.9	18.08(40)	0.02
2	2	6.7	23.24(47)	–
2	3	1.6	21.88(47)	–

4.5 Amplitude-dependent mechanical damping

For spring 2, the run-to-run scatter in γ is not random. It correlates almost perfectly with the initial EMF amplitude ($r = 0.98$, $p < 10^{-4}$; Fig. 4). The initial EMF amplitude is a proxy for the release amplitude. A larger release gives faster decay, which is the signature of a nonlinear loss such as quadratic air drag on the cradle or amplitude-dependent friction at the suspension. Spring 1 shows no such correlation ($r = -0.06$), and its release amplitude also varied less. Several video tracks for spring 2 show a slow beating in the envelope, which suggests energy exchange with a pendulum or torsional mode of the cradle. That would also make the effective damping depend on how the motion was started.

5 Discussion

Why the resistance did not matter. In the DAQ, the coil connects between the virtual ground and the high-impedance non-inverting input of U2B. The only DC path closing the coil circuit is the $1\,\text{M}\Omega$ bias resistor R5 (Fig. 1). Placing R_{ext} in series with the coil therefore changed R_{loop} from about $1.000\,\text{M}\Omega$ to at most $1.010\,\text{M}\Omega$. That is a 1% change, in a loop whose current was already limited to about $16\,\text{mV}/1\,\text{M}\Omega \approx 16\,\text{nA}$. By Eq. (6), the electromagnetic damping in every run was $\gamma_{\text{em}} \approx 2 \times 10^{-8}\,\text{s}^{-1}$, about one millionth of the measured damping. The experiment measured the

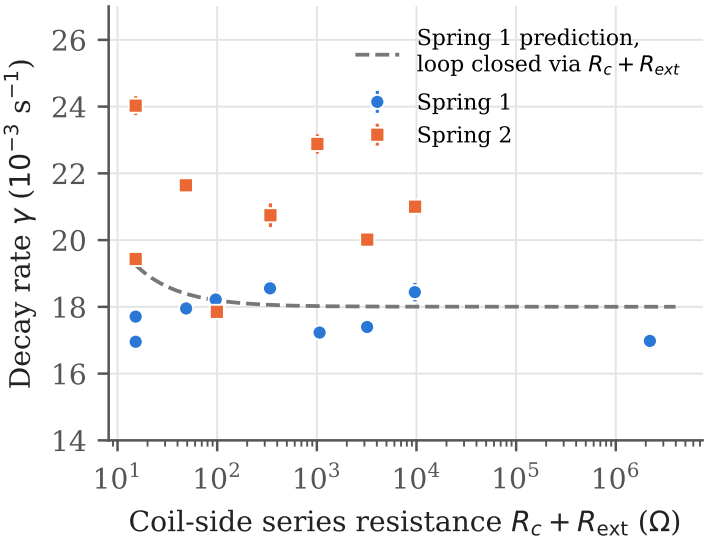


Figure 3: Decay rate against coil-side series resistance. The dashed curve is the prediction of Eq. (6) for spring 1 if the coil loop had been closed through $R_c + R_{\text{ext}}$. The prediction for the circuit as actually built, closed through 1 MΩ, is flat to within $2 \times 10^{-8}\,\text{s}^{-1}$. Error bars are fit standard errors and are mostly smaller than the markers.

open-circuit EMF accurately, but no braking current ever flowed.

The 2.2 MΩ run independently supports this picture. With R5 completing the loop, R_{ext} and R5 form a voltage divider, so the measured EMF should drop to $10^6/(3.2 \times 10^6) = 0.31$ of its usual value. We observe 0.25 ± 0.02 (scatter of the other runs: 6%). The order of magnitude agrees, and the remaining 20% is plausibly due to additional loading at the input, which we have not characterised.

How to measure electromagnetic damping with this DAQ. The braking current must flow through a known, low resistance, and the DAQ should measure the voltage across it. In practice, place R_{load} directly across the coil terminals, in parallel with the DAQ input, so that $R_{\text{loop}} = R_c + R_{\text{load}}$, and vary R_{load} from 0 Ω (shorted coil) upward. The DAQ then reads $\varepsilon R_{\text{load}}/(R_c + R_{\text{load}})$. A shorted-coil run cannot be measured this way, but it can be compared with an open-coil run to give γ_{em} directly from the difference in decay rate.

Even with the correct circuit, our coil 1 would give at most a 7% effect. A convincing $1/R$ curve needs a larger coupling-to-mechanical-loss ratio. Options include more turns of thicker wire, setting the magnet's equilibrium at $z \approx a/2$ where $|\mathrm{d}\Phi/\mathrm{d}z|$ is largest, a stronger single magnet, and a rigid magnet mount in place of the string cradle to reduce mechanical losses and suppress the parasitic modes.

Spring comparison. Spring 2 damps faster than spring 1 ($\gamma = 20.9$ vs. $17.8 \times 10^{-3}\,\text{s}^{-1}$). Since the coupling to the coil is negligible, this difference is mechanical. With only two springs, no functional dependence of b on k can be inferred. The internship report's linear fit through two points, and the relation $b \propto A/(Rk)$ proposed there, are not supported by the data.

An unresolved inconsistency. With $m = 105\,\text{g}$, the static spring constants predict $f = \sqrt{k/m}/2\pi = 2.00\,\text{Hz}$ (spring 1) and 2.17 Hz (spring 2). The measured frequencies are 2.633 Hz and 2.333 Hz, which are higher and in the opposite order. Either the k values or the spring labels need re-checking, or the effective oscillating mass is smaller than 105 g [resolve before submission]. None of our conclusions depend on k, because γ and Q come from the time series alone. Absolute values of $b = 2m\gamma$, however, scale with the mass assumed.

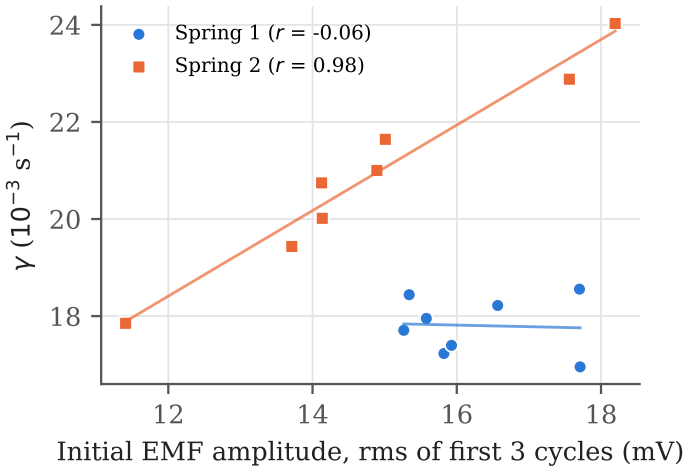


Figure 4: Decay rate against the rms EMF of the first three cycles, a proxy for release amplitude (coil 1, $R_{\text{ext}} < 10\,\text{k}\Omega$). Lines are least-squares fits.

Video tracking. The tracking-derived decay rates (Table 1) scatter more than the EMF-derived ones and are systematically lower. The tracks were limited by motion blur, so we treat them as qualitative confirmation of the oscillation amplitude and of the beating in spring 2, not as an independent measurement of γ.

Limitations. Each resistance was recorded once (twice at $0\,\Omega$), the release amplitude was not controlled well enough for spring 2, the coil geometries were not documented, and the amplitude A used in Eq. (6) comes from the video tracks. These limits affect the size of the predicted effect, but not the central conclusion, which rests on a 10^6-fold suppression.

6 Conclusion

A low-cost DAQ (US$43) measured millivolt-level coil EMF with a $34\,\mu\text{V}$ noise floor and determined the decay rate of a magnet–spring oscillator to $\approx 1\%$ per run. Changing the series resistance from $15\,\Omega$ to $2.2\,\text{M}\Omega$ did not change the damping. A model built only from the measured EMF predicts a 7% effect for a loop closed through the coil, and the data exclude it at 3.7σ. The reason is that the coil circuit was closed only through the DAQ's $1\,\text{M}\Omega$ bias resistor, which suppressed the electromagnetic damping by a factor of about 10^6. The damping we measured is mechanical, and for one spring it is strongly amplitude-dependent. We give a corrected circuit topology and design guidance for observing the expected $1/R_{\text{loop}}$ dependence.

Data and code availability

Schematics, firmware, all raw data files and the analysis scripts used for this paper are available at `https://github.com/Umerr7/electromagnetic-damping-daq` [11] under the MIT licence [add the `analysis/` scripts from this paper to the repository and archive a release on Zenodo for a DOI].

Author contributions

[Confirm before submission.] H.T.M.: conceptualisation, theory, formal analysis, writing (original draft); H.K.: analysis software (plotting and Butterworth filtering), video tracking; M.U.F.: DAQ hardware and firmware design, data collection; M.S.Q., S.M., P.H.: supervision. All authors reviewed the manuscript.

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Use of AI tools

An AI assistant (Claude, Anthropic) was used to reanalyse the raw data with the scripts in the repository and to restructure and edit the manuscript. The authors checked all analysis and text and take full responsibility for the content.

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